

REAL ESTATE MARKET TRENDS

Property Market Dashboard

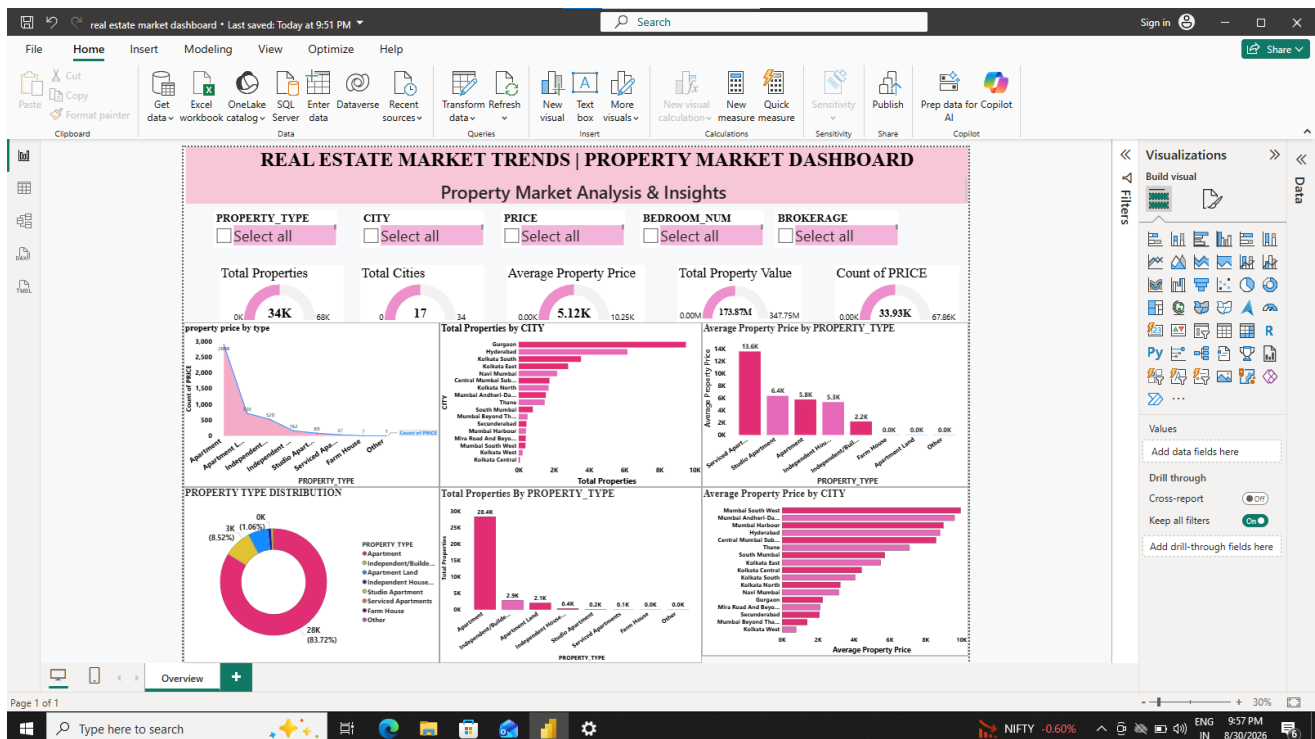
Project Report

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1. Introduction

Real estate generates large amounts of property information related to property type, location, price, bedrooms and brokerage. Examining these records as raw data can make it difficult to identify market patterns quickly. This project uses Microsoft Power BI to transform real estate data into an interactive dashboard for property-market analysis and decision support.

2. Problem Statement

The main problem addressed by this project is the difficulty of obtaining a clear and interactive view of important real estate metrics from property-level data. The dashboard is designed to help users compare property volumes, prices, property types and city-level patterns through visual analytics.

3. Objectives

- Build an interactive real estate market dashboard using Power BI.
- Track important property-market KPIs.
- Analyze property distribution by property type.
- Compare property volume across cities.
- Compare average property prices across property types and cities.
- Allow users to filter the dashboard using property type, city, price, bedroom number and brokerage.
- Present market insights in a simple and decision-friendly format.

4. Tools and Technologies

Tool / Technology	Purpose
Microsoft Power BI Desktop	Data transformation, modeling, DAX, visualizations and dashboard development.
Power Query	Data preparation and transformation.
DAX	Analytical calculations and KPI measures.
Charts and Slicers	Interactive market analysis and visual reporting.
Real Estate Dataset	Source data for property-market analysis.

5. Dashboard KPIs

KPI	Displayed Value
Total Properties	≈ 34K
Total Cities	17
Average Property Price	≈ 5.12K
Total Property Value	≈ 173.87M
Count of PRICE	≈ 33.93K

6. Interactive Slicers

- PROPERTY_TYPE – filters the dashboard by selected property category.
- CITY – filters the analysis by selected city/locality.
- PRICE – supports focused analysis of selected price ranges.
- BEDROOM_NUM – enables comparison by bedroom count.
- BROKERAGE – allows brokerage-related filtering.

7. Data Analysis and Visualizations

Visualization	Explanation
Property price by property type	Shows the count of price records across different property categories.
Total Property by CITY	Compares and ranks cities according to the number of property records.
Average Property Price by PROPERTY_TYPE	Compares average property prices across property categories.
Property Type Distribution	Donut chart showing the share of properties represented by each property type.
Total Properties by PROPERTY_TYPE	Compares property counts across categories.
Average Property Price by CITY	Ranks cities/localities according to average property price.

8. Dashboard Design

The dashboard uses a clean single-page layout with a prominent title, a dedicated filter section, KPI cards and a structured grid of analytical visuals. The upper section gives a quick market summary, while the lower charts provide detailed comparisons. The pink and white visual theme creates a consistent presentation.

9. Key Observations

- The dashboard represents approximately 34K properties across 17 cities.
- Apartments are the dominant property category, accounting for about 83.72% of the displayed distribution.
- Gurgaon and Hyderabad appear among the highest-volume cities in the displayed property-count ranking.
- Service Apartments show the highest displayed average property price at approximately 13.6K.
- Several Mumbai localities, including Mumbai South West and Mumbai Andheri-Dadar, appear near the top of the city average-price ranking.
- The dashboard enables users to examine these patterns interactively through multiple slicers.

10. Methodology

Data Preparation

Relevant real estate attributes were prepared for analysis so that property categories, locations, prices and other fields could be used consistently.

Data Transformation

Power Query can be used to clean, format and structure the source data before visualization.

Data Modeling

The prepared dataset was organized for analysis through Power BI's data model.

DAX Measures

Measures were used for key metrics such as property count, city count, average price and total property value.

Visualization

Charts, KPI cards and slicers were arranged into a single interactive dashboard.

Analysis

The final dashboard was interpreted to identify property-type concentration, high-volume markets and high-price locations.

11. Business Insights

- Apartment-focused supply: The strong share of apartments indicates that this category has a major influence on the overall dataset.
- High-volume locations: Gurgaon and Hyderabad are important markets in terms of the displayed number of property records.
- Premium property category: Service Apartments have the highest displayed average price among the visible property types.
- Mumbai price concentration: Several Mumbai areas rank highly in average property price, indicating higher displayed price levels in these localities.
- Interactive segmentation: Filters make it possible to study smaller market segments instead of relying only on overall averages.

12. Advantages

- Interactive and easy-to-understand reporting.
- Multiple KPIs and charts are available on one page.
- Supports city and property-type comparison.
- Reduces the effort required to interpret large property datasets.
- Provides a strong foundation for business intelligence and real estate analysis.

13. Limitations

- The report is based on the dashboard information visible in the supplied project snapshot.
- The screenshot does not explicitly identify the currency/unit used for prices.
- The dashboard is primarily descriptive and does not provide future price forecasting.
- Low-volume categories should be interpreted carefully.

14. Future Scope

- Add time-based property price and volume trends if a date field is available.
- Add price-per-square-foot analysis if area information is available.
- Add map-based geographic analysis.
- Add bedroom-wise price and property-type comparisons.
- Add brokerage performance analysis.
- Add predictive analytics or machine-learning-based price forecasting.
- Connect the dashboard to a regularly refreshed data source.

15. Conclusion

The Real Estate Market Trends | Property Market Dashboard demonstrates the practical use of Microsoft Power BI for transforming property data into an interactive business intelligence report. The dashboard brings together market-size KPIs, property-type distribution, city-level property volumes and average price comparisons in one visual interface.

The analysis shows a strong dominance of apartments in the displayed dataset, significant property volumes in markets such as Gurgaon and Hyderabad, a higher average displayed price for Service Apartments, and comparatively high average prices across several Mumbai localities. The project therefore provides a useful foundation for understanding property-market patterns and demonstrates skills in data analysis, Power BI visualization and analytical storytelling.

16. Internship Learning Outcomes

- Learned how business intelligence dashboards are structured.
- Developed practical experience with Power BI visuals and slicers.
- Improved understanding of KPI-based reporting.
- Learned to compare categorical and geographic market segments.
- Improved data storytelling and presentation skills.
- Gained practical exposure to applying analytics concepts to a real-world business domain.

17. Project Evidence

Figure 1: Real Estate Market Trends | Property Market Dashboard developed for the CodeAlpha internship.

